Causes of Hallucination in Large Language Models

A hallucination in a large language model is a fluent answer that is unsupported, inaccurate, or inconsistent with the available information. The main point of this document is causal explanation: why a model can produce such text even when the answer sounds confident.

One cause is probabilistic generation. A language model predicts likely next tokens from patterns learned during training. It does not automatically verify each statement against a trusted source. When the prompt is vague or the context is incomplete, the model may fill gaps with plausible wording.

A second cause is incomplete or noisy training data. If facts in the data are outdated, contradictory, or weakly represented, the model can combine fragments from different contexts. This can lead to wrong names, wrong dates, invented relationships, or fabricated citations.

A third cause is weak grounding at inference time. If a question requires a specific source but no evidence is supplied, the model may answer from memory. In a retrieval-augmented system, grounding can still fail when the retriever returns only related passages instead of passages that contain the exact fact.

Other causes include overgeneralization, prompt ambiguity, exposure bias, and decoding choices. High temperature sampling can make outputs more diverse but also less stable. Long answers may accumulate more unsupported details because each extra sentence introduces another factual claim.

Mitigation usually combines better retrieval, clear prompting, claim-level checking, conservative correction, and refusal when the evidence does not answer the question. The safest system distinguishes between an answer that is supported by evidence and an answer that merely sounds relevant.